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Investigating biological sex as a moderator of the association of nature exposure with brain health: a cross-sectional UK biobank analysis

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In this cross-sectional analysis of the UK Biobank, we investigated whether biological sex moderates the association of residential nature exposure with brain volume or cognitive function. We included 11,448 cognitively healthy UK residents aged 37–73 years (98% White; 51% female). Residential nature exposure was estimated as the percentage of land classified as natural environment within 1000 m and 300 m buffers around each participant's home. Structural brain magnetic resonance imaging (MRI) outcomes included total grey matter volume, total white matter volume, and average hippocampal volume, all normalized for head size. Cognitive function was assessed with the Trail Making Test (B-A) and the Symbol Digit Substituton Test. In linear regression models, higher residential nature exposure at both buffer sizes was associated with greater grey matter volume (1000 m: β = 629 mm^3 per 10% increment in nature exposure; 95% CI: 234 to 1023; $p = 0.002$; 300 m: β = 642 mm^3; 95% CI: 286 to 997; $p < 0.001$), greater white matter volume (1000 m: 659 mm^3; 95% CI: 229 to 1089; $p = 0.003$; 300 m: β = 527 mm^3; 95% CI: 140 to 914; $p = 0.008$), and more correct matches on the Symbol Digit Substituton Test (1000 m: β = 0.106 matches; 95% CI: 0.057 to 0.154; $p < 0.001$; 300 m: β = 0.049 matches; 95% CI: 0.006 to 0.092; $p = 0.03$). In males, compared with females, higher nature exposure was associated with a greater increment in grey matter volume (1000 m; β = 635 mm^3; 95% CI: 53 to 1217; $p = 0.03$; 300 m: β = 634 mm^3; 95% CI: 102 to 1167; $p = 0.02$), and with a greater reduction in Trail-Making Test B-A time (300 m only: β = 0.284 s; 95% CI: 0.016 to 0.551; $p = 0.04$). These sex differences showed some sensitivity to participant residence changes and to type of nature exposure measure. Residential nature exposure may support brain volume and cognitive function, and some of the potential benefits may vary by sex.

Dementia is a major global healthcare burden[1]. With no effective pharmaceutical treatment widely available, the World Health Organization urges the development of low-cost, accessible strategies to prevent or delay the onset and progression of cognitive impairment and dementia[2].

Nature spaces, like parks and forests, are a community-level feature that may support brain health and reduce the risk of cognitive decline[3]. Observational evidence suggests that, among adults, residential nature exposure is associated with greater regional grey matter volume[4], cortical thickness[5–7], and structural integrity[8], as well as better cognitive performance across multiple domains[5,9–17].

Despite these promising findings, there are key limitations to our knowledge about the relationship of nature exposure with brain and cognitive health. Most prior studies have focused on greenspace, i.e., outdoor land covered in green vegetation[18,19], rather than all types of nature spaces (e.g., land covered by vegetation,

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dirt, rock, or water)[20,21]. These studies may therefore have underrepresented nature exposure[22], leading to less accurate estimates of nature-health associations. Moreover, few studies have examined potential moderators of the association of nature exposure with brain or cognitive health. Identifying moderators would help inform on the conditions and the individual characteristics under which nature exposure may be beneficial[3].

Biological sex is a moderator of interest because there are notable sex differences in dementia risk and in brain and cognitive aging[23–27]. Females have double the prevalence of Alzheimer's disease (AD) versus males[28], show greater brain atrophy than males after onset of mild cognitive impairment or AD[24], and experience greater β-amyloid deposition and hippocampal atrophy during the menopausal transition, as compared with age-matched males[25]. Moreover, various modifiable factors are hypothesized to link nature exposure to brain or cognitive health—such as stress, mental health, physical activity, and air pollution[22]—and there are sex differences within many of these pathways[22,2329–35]. For example, females are more vulnerable to stress-associated cognitive decline or impairment[32,36,37], and nature exposure can reduce stress[38–40], potentially in a sex-dependent manner[33–35]. Females may therefore be more responsive to brain or cognitive benefits of nature that occur through stress reduction. Given these potentially interacting pathways, it is plausible that sex could moderate the association of nature exposure with brain volume or with cognitive function.

Few studies have examined whether biological sex moderates the association of nature exposure with brain volume or with cognitive function, and the results have been equivocal[7,10,11,16,41]. For example, Besser et al. (2021) did not find sex differences in the association of residential greenspace with brain structure[41], while De Keijzer et al. (2018) found that residential greenspace was associated with slower global cognitive decline in females than in males[16]. Among these studies, most had small samples ($N < 300$),[7,10,11] and/or conducted sex-stratified analyses without formally testing moderation by sex[7,10,11,16]. Given the potential of nature exposure in promoting brain and cognitive health[3], and that nearly 2/3 of dementia cases are among females[28], more research—using more comprehensive measures of nature exposure, large samples, and formal tests of moderation—is warranted to clarify if there are sex differences in the association of nature exposure with brain and cognitive health.

To address these gaps, using data from the UK Biobank, we assessed the association of nature exposure with brain volume or with cognitive function, and potential sex differences therein.

Methods

Participants

The UK Biobank (http://www.ukbiobank.ac.uk) is a prospective cohort with genetic, lifestyle, and health data on over 502,000 UK residents[42]. From 2006 to 2010, individuals aged 37–73 years and registered with the National Health Service (NHS) were recruited by mail to one of 22 assessment centres throughout England, Wales, and Scotland for a baseline assessment, at which time nature exposure data were documented. From 2014 to 2015, all 340,000 participants with an email address were invited by email to complete an online, self-administered neuropsychological battery; ~120,000 participants completed all online neuropsychological tests. From 2014 onward, ~50,000 participants completed structural brain magnetic resonance imaging (MRI) scans at one of three imaging centres.

Standard protocol approvals, registrations, and consents

Ethics approval for the UK Biobank study was provided by the NHS National Research Ethics Service (11/NW/0382). All UK Biobank participants gave written informed consent prior to participation. All research was performed in accordance with relevant guidelines and regulations. The current analyses were conducted on data extracted October 7, 2021 as part of UK Biobank Application 69,022.

Eligibility criteria

We included participants with complete data for nature exposure (collected at baseline, 2006–2010), structural brain MRI (completed 2014–2020), neuropsychological testing (completed 2014–2015), and covariates (collected at baseline, 2006–2010). We excluded participants with a diagnosis of dementia, Parkinson's disease, or stroke.

Measures

Nature exposure

Residential nature exposure was estimated as the percentage of land cover classified as "natural environment"—as opposed to "built environment"—in 1000-m and 300-m buffers around participants' home locations. Land cover data were obtained from the UK Centre for Ecology and Hydrology (CEH) 2007 Land Cover Map (LCM)[43]. Given the variable findings and uncertainty regarding buffer sizes relevant to nature-health associations[22,44], we used both the 1000-m and 300-m buffers to examine the broader neighbourhood and the immediate residential area, respectively, as potentially relevant geographic contexts[44,45]. We chose the land cover-based "natural environment" measure for our primary analysis, rather than the land use-based "greenspace" measure, because it includes a broader set of natural environment types beyond just greenspace (e.g., water and rock areas). Sensitivity analyses with the alternative nature exposure measure ("greenspace") are described below.

Structural brain MRI

The UK Biobank MRI acquisition protocol[46] and processing pipeline[47] have been described previously and are documented on the UK Biobank website[48]. Structural brain MRI scans were acquired on a Siemens Skyra 3 T scanner, with a standard 32-channel RF receive head coil[46]. Brain volumes were generated from T1-weighted structural images by the image-processing pipeline developed by and run on behalf of the UK Biobank[47]. Our study outcomes included total grey matter volume, total white matter volume, and hippocampal volume (averaged across both hemispheres). Each brain volume variable was normalized for head size. Total grey and white matter volumes were normalised for head size by the UK Biobank using a T1-based estimate of the external

surface of the skull[47]. To normalise hippocampal volume, which was provided as a raw value, we multiplied the raw volume by the T1-based head size scaling factor provided by the UK Biobank (data field 25000).

Cognitive function

UK Biobank participants completed an online neuropsychological testing battery from 2014 to 2015, detailed on the UK Biobank website[49]. Among these tests, we examined the Trail Making Test (TMT) versions A and B and the Symbol Digit Substituton Test because they are sensitive to age-related decline and changes in cognitive status[50–52], and associated with daily function in older adults with and without cognitive impairment[53,54]. Additionally, a systematic review of acute experimental studies showed that executive functions were benefitted by exposure to natural environments, compared with built environments[55].

In TMT A, participants were presented a series of circles labelled with numbers, and instructed to connect them in ascending order. In TMT B, participants were presented a series of circles labelled with numbers or letters, and instructed to connect them in ascending order, alternating between numbers and letters[56]. We indexed the difference in time to complete versions B and A (i.e., TMT B-A) as a measure of set-shifting, with a smaller time difference indicating better performance[51].

The Symbol Digit Substituton Test measures complex processing speed[57]. Participants were presented a sequence of randomly arranged symbols, and instructed to match each symbol to a number according to a code provided. The total number of correct matches made in one minute was the score, with a higher number indicating better complex processing speed[57].

Moderator

Biological sex was recorded as male or female, acquired from the NHS central registry at recruitment.

Covariates

We identified ten covariates that have previously been associated with brain and cognitive health[58–60]. Age at baseline was derived from date of birth and truncated to whole years. BMI was calculated from height and weight measured in-person. Participants self-reported their educational attainment, average annual total household income, time spent outdoors, smoking status, and disability status. Physical activity was self-reported using the International Physical Activity Questionnaire (IPAQ) short form[61]. We estimated neighbourhood-level socioeconomic status with the Index of Multiple Deprivation[62]. Home area population density was determined by the UK Biobank from participant postal codes at the time of recruitment matched to 2001 census data.

Other descriptive variables

To help characterize our sample, we also included the following descriptive variables, all of which were self-reported by participants through a digital questionnaire: ethnic background, hearing difficulties, diabetes diagnosed by a doctor, and cardiovascular problems diagnosed by a doctor (including heart attack, angina, stroke, and hypertension).

Statistical analysis

We performed all analyses in R (version 4.2.3) using RStudio (version 2022.02.3). Descriptive statistics for demographic and outcome variables were calculated, wherein we tested differences between sexes using independent samples t-tests for continuous variables and chi-squared tests for categorical variables.

We conducted linear regressions to assess the independent associations of residential nature exposure with brain volume or cognitive function. Separate models for each brain and cognitive outcome were developed, adjusting for all covariates. Crude models, adjusted only for age, sex, and ethnic background, were also developed (Supplementary Tables S1 and S2).

We then assessed whether sex moderated the associations of residential nature exposure with brain volume or cognitive function by adding an interaction term of nature exposure*sex to all regression models. For outcomes with significant interactions, we conducted post-hoc, sex-stratified regressions to evaluate the association of residential nature exposure with the outcome within each sex.

We conducted two sensitivity analyses. First, as an alternative nature exposure measure, we used the percentage of land classified as "greenspace", among 10 land-use categories, in 1000-m and 300-m buffers around participants' home locations. Land use data were obtained from the 2005 Generalized Land Use Database for England (GLUD) for 2001 Census Output Areas, and were therefore limited to participants residing in England[63]. We conducted these analyses because reviews of observational studies on nature-health associations indicate that the nature exposure measure can moderate the strength and significance of associations, and thus recommend the use of multiple exposure measures[22,44,64]. Secondly, to account for possible changes in residence between the time of data collection for nature exposure (baseline; 2006–2010) and the time of the MRI and cognitive assessments (follow-up; 2014–2015), we repeated all analyses while restricting the sample to participants whose residence remained unchanged between the baseline and follow-up assessments. To do this, we excluded participants whose time at current address was less than the time between baseline and follow-up assessments.

All analyses were exploratory; the overall alpha was < 0.05.

Results

Participant characteristics

From the parent sample of 502,422 UK Biobank participants, our final analytic sample included 11,448 individuals (Fig. 1). Our sample had a mean age of 55.0 years (SD = 7.5 years), 51% were female ($n = 5{,}784$), and 98% were White.

Table 1 shows sex-stratified sample characteristics. Males were older, spent more time outdoors, had a higher prevalence of hearing difficulty, diabetes, and cardiovascular problems, and had a lower prevalence of probable major depressive disorder. Females had lower residential natural environment and greenspace percentage (buffered to 1000 m and 300 m), better scores on the Trail-Making Test B-A and the Symbol Digit Substituton Test, greater total grey matter volume (normalized for head size) and average hippocampal volume (normalized for head size), and lower total white matter volume (normalized for head size).

Associations of residential nature exposure with brain volume or cognitive function

At both the 1000 m and 300 m buffers, every 10% increment in residential nature exposure was associated with greater grey matter volume (1000 m: $\beta = 629$ mm^3; 95% CI: 234 to 1023; $p = 0.002$; 300 m: $\beta = 642$ mm^3; 95% CI: 286 to 997; $p < 0.001$), greater white matter volume (1000 m: 659 mm^3; 95% CI: 229 to 1089; $p = 0.003$; 300 m: $\beta = 527$ mm^3; 95% CI: 140 to 914; $p = 0.008$), and more correct matches on the Symbol Digit Substituton Test (1000 m: $\beta = 0.106$ matches; 95% CI: 0.057 to 0.154; $p < 0.001$; 300 m: $\beta = 0.049$ matches; 95% CI: 0.006 to 0.092; $p = 0.03$; Table 2). Residential nature exposure was not associated with average hippocampal volume (1000 m: $\beta = 0.39$ mm^3 per 10% increment in nature exposure; 95% CI: −5.47 to 6.25; $p = 0.90$; 300 m: $\beta = -3.76$mm^3; 95% CI:

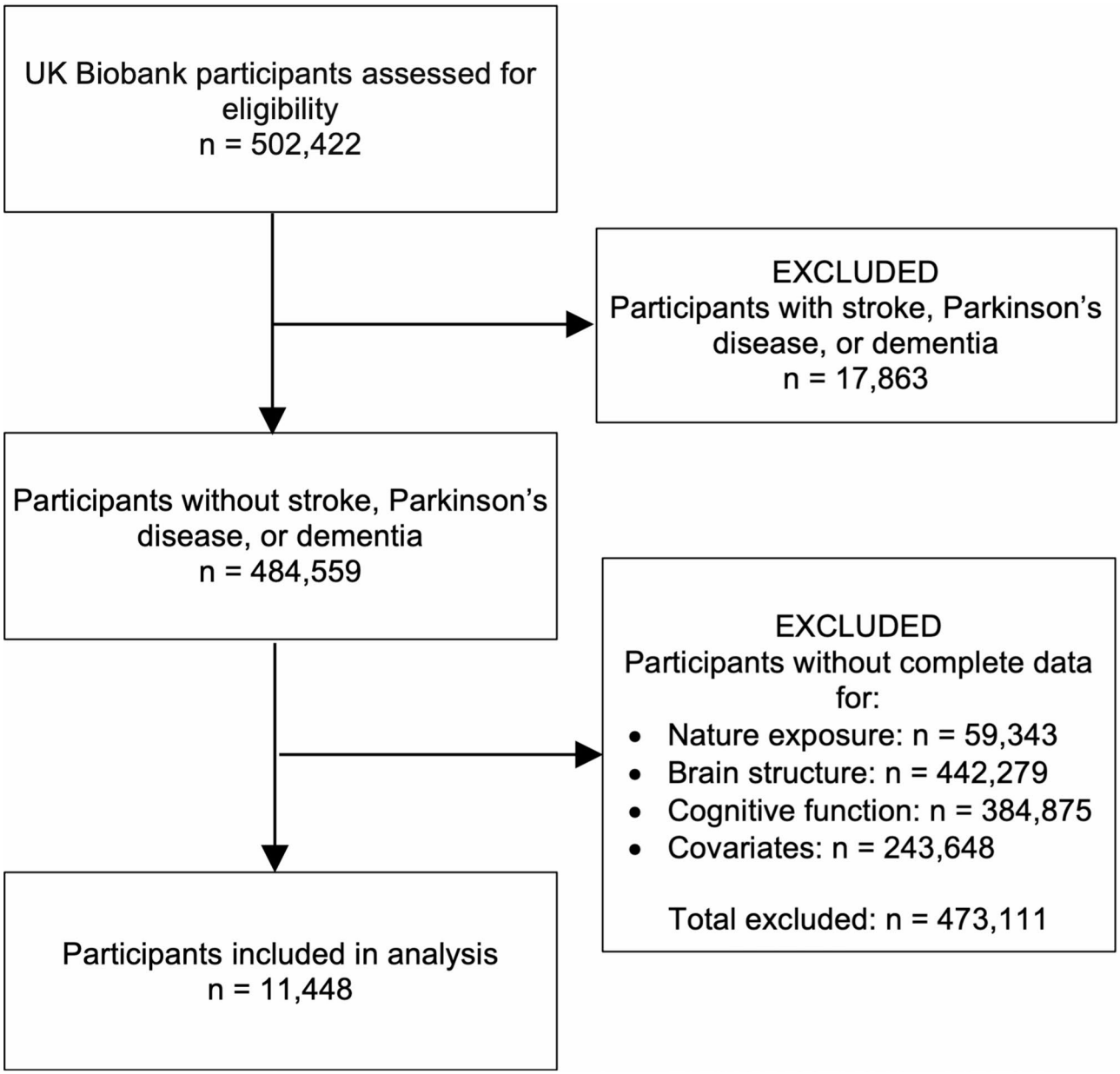

Fig. 1. Participant flow for the current cross-sectional analysis of the UK Biobank. Note that the total number of participants excluded for missing data accounts for overlap, i.e.. participants with missing data across more than one of the four categories of variables (nature exposure, brain structure, cognitive function, or covariates).

Characteristic	Female *N* = 5,784	Male *N* = 5,664	*p*-value
Age (years)	53.84 (7.21)	56.08 (7.54)	< 0.001
Body mass index (kg/m^2)	25.93 (4.56)	26.84 (3.78)	< 0.001
Ethnic background			0.13
White	5,645 (98%)	5,526 (98%)	
Asian or Asian British	33 (0.6%)	48 (0.9%)	
Black or Black British	28 (0.5%)	24 (0.4%)	
Mixed	25 (0.4%)	15 (0.3%)	
Unknown	53 (0.9%)	51 (0.9%)	
Education (n, %)			< 0.001
College or university degree	2,947 (51%)	3,073 (54%)	
A levels/AS levels or equivalent	876 (15%)	698 (12%)	
O levels/GCSEs or equivalent	1,147 (20%)	936 (17%)	
CSEs or equivalent	173 (3.0%)	180 (3.2%)	
NVQ/HND/HNC or equivalent	155 (2.7%)	371 (6.6%)	
Other professional qualifications	310 (5.4%)	177 (3.1%)	
None of the above	176 (3.0%)	229 (4.0%)	
Household income (GBP)			< 0.001
<18,000	694 (12%)	507 (9.0%)	
18,000–30,999	1,375 (24%)	1,096 (19%)	
31,000–51,999	1,733 (30%)	1,766 (31%)	
52,000–100,000	1,511 (26%)	1,794 (32%)	
>100,000	471 (8.1%)	501 (8.8%)	
Index of Multiple Deprivation	17.80 (14.39)	18.14 (14.16)	0.2
Home area population density			0.6
Urban area	4,819 (83%)	4,690 (83%)	
Town	445 (7.7%)	464 (8.2%)	
Village, Hamlet, or Rural Area	520 (9.0%)	510 (9.0%)	
Self-reported physical activity (min/day)	118.56 (87.26)	119.63 (90.64)	0.5
Average time outdoors (hours/day)	2.18 (1.37)	2.58 (1.75)	< 0.001
Smoking status			< 0.001
Current smoker	297 (5.1%)	384 (6.8%)	
Previous smoker	1,820 (31%)	2,038 (36%)	
Never	3,667 (63%)	3,242 (57%)	
Living with illness, disability, or infirmity (n, %)	1,203 (21%)	1,420 (25%)	< 0.001
Hearing problems (n, %)	1,110 (19%)	1,501 (27%)	< 0.001
Living with cardiovascular disease (n, %)	878 (15%)	1,397 (25%)	< 0.001
Living with diabetes mellitus (n, %)	80 (1.4%)	169 (3.0%)	< 0.001
Probable Recurrent Major Depressive Disorder (n, %)	426 (27%)	242 (15%)	< 0.001
Natural environment percentage (buffered to 1000 m)	47.03 (22.45)	48.01 (22.11)	0.02
Natural environment percentage (buffered to 300 m)	37.65 (24.49)	37.99 (24.18)	0.5
Greenspace percentage (buffered to 1000 m)	43.68 (26.24)	44.94 (25.85)	0.007
Greenspace percentage (buffered to 300 m)	29.06 (26.66)	29.90 (26.52)	0.094
Trail-Making Test (B-A; sec)	25.11 (18.79)	26.02 (17.95)	0.009
Symbol Digit Substitution Test (number of correct matches)	20.91 (5.01)	20.42 (4.58)	< 0.001
Total grey matter volume (head size-adjusted; cm^3)	808.45 (45.43)	775.19 (44.42)	< 0.001
Total white matter volume (head size-adjusted; cm^3)	699.25 (40.30)	704.74 (40.69)	< 0.001
Average hippocampal volume (head size-adjusted; mm^3)	5,108.73 (522.77)	4,776.51 (575.74)	< 0.001

Table 1. Descriptive statistics and outcomes of UK biobank participants included in the current analysis, stratified by sex. For continuous variables, data presented are mean (SD) for each sex and p-values for independent samples t-tests comparing means between sexes. For categorical variables, data presented are n (%) within each sex and p-values for chi-square tests comparing distributions between sexes. Acronyms: A levels: General Certificate of Education Advanced Level; AS levels: General Certificate of Education Advanced Subsidiary Level; O levels: General Certificate of Education Ordinary Level; GCSE: General Certificate of Secondary Education; CSE: Certificate of Secondary Education; NVQ: National Vocational Qualification; HND: Higher National Diploma; HNC: Higher National Certificate.

	1000 m buffer (*N* = 11,448)		300 m buffer (*N* = 11,448)	
Outcome	**β (95% CI)**	**p-value**	**β (95% CI)**	**p-value**
Brain volume				
Total grey matter volume (head size-adjusted; mm^3)	629 (234, 1023)	0.002	642 (286, 997)	< 0.001
Total white matter volume (head size-adjusted; mm^3)	659 (229, 1089)	0.003	527 (140, 914)	0.008
Average hippocampal volume (head size-adjusted; mm^3)	0.39 (−5.47, 6.25)	0.90	−3.76 (−9.04, 0.00)	0.16
Cognitive function				
Trail-Making Test (B-A; sec)	−0.197 (−0.395, 0.001)	0.05	−0.120 (−0.299, 0.058)	0.19
Symbol Digit Substituton Test (number of correct matches)	0.106 (0.057, 0.154)	< 0.001	0.049 (0.006, 0.092)	0.03

Table 2. Results of linear regressions examining the association of residential nature exposure with brain volume or cognitive function outcomes in the UK biobank. Residential nature exposure was estimated as the proportion of land cover classified as "natural environment" within 1000-m or 300-m buffers. β represents the increment in the outcome associated with a 10% increment in nature exposure. Models were adjusted for age, sex, body mass index, education, income, physical activity, time spent outdoors, smoking status, disability status, Index of Mutliple Deprivation, and population density.

	1000 m buffer (*N* = 11,448)		300 m buffer (*N* = 11,448)	
Outcome	**β (95% CI)**	**p-value**	**Outcome**	**β (95% CI)**
Brain volume				
Total grey matter volume (head size-adjusted; mm^3)	−635 (−1217, −53)	0.03	−634 (−1167, 102)	0.02
Total white matter volume (head size-adjusted; mm^3)	−302 (−936, 332)	0.35	−291 (−872, 289)	0.32
Average hippocampal volume (head size-adjusted; mm^3)	−1.72 (−10.36, 6.92)	0.70	−5.90 (−13.81 to 2.02)	0.14
Cognitive function				
Trail-Making Test (B-A; sec)	0.144 (−0.148, 0.436)	0.33	0.284 (0.016, 0.551)	0.04
Symbol Digit Substituton Test (number of correct matches)	0.025 (−0.046, 0.096)	0.49	−0.002 (−0.067, 0.063)	0.96

Table 3. Results of linear regressions examining the moderating effect of biological sex on the association of residential nature exposure with brain volume or cognitive function in the UK biobank. Residential nature exposure was estimated as the proportion of land cover classified as "natural environment" within 1000-m or 300-m buffers. β represents the difference between sexes (female as reference) in the increment in the outcome associated with a 10% increment in residential nature exposure. Models were adjusted for age, body mass index, education, income, physical activity, time spent outdoors, smoking status, disability status, Index of Mutliple Deprivation, and population density.

−9.04 to 0.00; $p = 0.16$), nor with the TMT B-A (1000 m: β=−0.197 s; 95% CI: −0.395 to 0.001, $p = 0.05$; 300 m: β=−0.120 s; 95% CI: −0.299 to 0.058, $p = 0.19$).

Crude models—adjusted only for age, sex, and ethnic background—replicated the results of the fully adjusted models described above (Supplementary Table S1).

Moderation by biological sex on the association of residential nature exposure with brain volume or cognitive function

Biological sex moderated the association of residential nature exposure with total grey matter volume, such that every 10% increment in nature exposure was associated with a greater increment in grey matter volume in males than in females, for both the 1000 m (β = 635 mm^3; 95% CI: 53 to 1217; $p = 0.03$) and 300 m buffers (β = 634 mm^3; 95% CI: 102 to 1167; $p = 0.02$; Table 3). In post-hoc, sex-stratified analyses, the association of residential nature exposure with grey matter volume was significant in males (1000 m: β = 797 mm^3 per 10% increment in nature exposure; 95% CI: 234 to 1359; $p = 0.006$; 300 m: β = 882 mm^3; 95% CI: 374 to 1391; $p < 0.001$), but not in females (1000 m: β = 444 mm^3; 95% CI: −108 to 996; $p = 0.11$; 300 m: 447 mm^3; 95% CI: −47 to 941; $p = 0.08$).

For the 300 m nature exposure buffer—but not for the 1000 m buffer—biological sex also moderated the association of residential nature exposure with TMT B-A (Table 3). Every 10% increment in nature exposure was associated with a 0.284 s greater decrement in time in males than in females (95% CI: 0.016 to 0.551; $p = 0.04$). In post-hoc, sex-stratified analyses, the association of residential nature exposure with TMT B-A was significant

only in males (β=−0.318 s per 10% increment in nature exposure; 95% CI: −0.564 to −0.073; $p = 0.01$; females: β = 0.064 s; 95% CI: −0.193 to 0.322; $p = 0.62$).

Biological sex did not moderate the association of residential nature exposure with other brain volume or cognitive function outcomes.

Crude models—adjusted only for age, sex, and ethnic background—replicated the results of the fully adjusted models described above (Supplementary Table S2).

Sensitivity analyses

When we used greenspace—rather than natural environment—as the nature exposure measure, all except two results remained the same as the main analyses (Supplementary Tables S3 and S4). The association of residential nature exposure (buffered to 300 m) with the Digit Symbol Substitution test remained in the same direction, but was no longer significant (β = 0.031 matches; 95% CI: −0.007 to 0.069; $p = 0.11$). For moderation by sex on the association of residential nature exposure (buffered to 300 m) with TMT B-A, the result also remained in the same direction, but was no longer significant (β = 0.159 s more improvement in time in males than in females per 10% increment in nature exposure; 95% CI: −0.086 to 0.403; $p = 0.20$).

When we restricted the sample to participants with no changes in residence between exposure (2006–2010) and outcome (2014–2020) assessments, results remained largely unchanged (Supplementary Tables S5 and S6). However, for moderation by sex on the association of residential nature exposure (buffered to 1000 m) with total grey matter volume, the result remained in the same direction, but was no longer significant (β = 641 mm^3 more in males than in females per 10% increment in nature exposure; 95% CI: −11 to 1292; $p = 0.05$).

Discussion

In this cross-sectional analysis of the UK Biobank, we found that residential nature exposure was associated with greater total grey matter volume and greater total white matter volume, and better complex processing speed. Sex moderated the association of residential nature exposure with total grey matter volume and with set-shifting, whereby the associations were stronger among males than among females.

Our findings of favourable associations between residential nature exposure and brain volume concur with prior cross-sectional studies in adults that found favourable associations of residential greenspace with regional grey matter volumes[4], regional and global cortical thickness[5–7], regional structural integrity[8], and ventricle burden[41]. Together with the current study, the evidence to date generally suggests that greater residential nature exposure is associated with better brain structure in adults.

While beyond the scope of the current study, three broad pathways have been proposed by which nature exposure may benefit brain health: (1) reducing stress, restoring attention, and supporting mental health; (2) mitigating harmful stimuli, such as air pollution and noise; and (3) facilitating health-related behaviours or capacities, such as physical activity, social engagement, sleep, and immune function[3,22,65,66]. These pathways have been investigated as mediators by a small number of observational studies, with unclear results[3]. Future longitudinal analyses are needed to verify current findings and to elucidate mediating pathways by which nature exposure may promote brain health.

We found an association of residential nature exposure with complex processing speed, but not with set-shifting. Specifically, every 10% increment in residential nature exposure was associated with 0.106 (1000 m buffer) or 0.031 (300 m buffer) more correct matches on the Symbol Digit Substituton Test. However, in the Swedish BioFINDER cohort study, the minimum clinically important difference (MCID) on the Symbol Digit Modalities Test was 3.5 matches for cognitively unimpaired older adults[67]. Thus, the nature exposure–processing speed association observed in our study may not be clinically meaningful. Our findings are comparable to those of previous nature exposure studies examining similar cognitive outcomes[3,9,14,15]. A study of 13,594 women aged 50–69 years in the USA showed an association of residential NDVI with a psychomotor speed/attention composite, measured by the Cogstate Brief Battery[9]. In two studies conducted in Europe ($n = 884$; Mental Alternation Test)[14] and in the USA ($n = 1602$; Colour Trails Test)[15] using tests similar to the Trail-Making Test B, there was a mix of favourable and null associations of cognitive performance with various measures of nature exposure. While methodological differences between studies —including differences in nature exposure measures, outcome measures, sample sizes and demographics, and study design (cross-sectional versus longitudinal)— limit comparability and likely contribute to much of the variation in findings[3,22,44], the potential benefit of nature exposure may vary by cognitive domain[3,55,68].

We found a stronger association of residential nature exposure with grey matter volume among males than among females, across both buffer sizes and both nature exposure measures. Residential natural environment, buffered to 300 m, was also associated with better set-shifting among males than among females, although the difference—a 0.284-sec greater improvement in TMT B-A score per 10% increment in residential nature exposure—was much less than the MCID of 12.7 s reported for cognitively unimpaired older adults[67]. Our findings contrasted somewhat with the minimal literature on sex differences in the association of nature exposure with brain or cognitive health. Two cross-sectional studies in middle-aged[7] and/or older adults[7,41] found no sex differences in the association of residential greenspace with brain structure[7,41], while two of three longitudinal studies found that residential greenspace was associated with slower cognitive decline in females but not in males[10,11,16]. Notably, only one of these five studies conducted a formal statistical test for moderation[41], whereas the remaining studies conducted sex-stratified analyses[7,10,11,16]. Based on the sex differences suggested in the present study, more studies are needed to assess moderation by sex with formal statistical testing.

Some of our findings, especially our moderation results, were sensitive to buffer size, nature exposure measure, and participant changes in residence. When the exposure measure was greenspace instead of natural environment, two results were no longer significant: the association of nature exposure, buffered to 300 m, with complex processing speed, and moderation by sex on the association of nature exposure, buffered to 300

m, with set-shifting. These results suggest that nature spaces beyond just green vegetation may also contribute to potential cognitive benefits, and reiterate previous findings of favourable associations varying across nature exposure measures and/or qualities of the nature itself[22]. When we excluded those with changes in residence, moderation by sex on the association of residential nature exposure with total grey matter volume remained robust for the 300 m buffer, but was no longer significant—although remained in the same direction—for the 1000 m buffer. This result emphasizes the importance of longitudinal tracking of residential address in tandem with outcomes when studying residential nature exposure.

Compared to the 1000 m nature exposure buffer, the 300 m buffer showed an additional moderation result for set-shifting and a more robust moderation result for grey matter volume. The 300 m buffer may be more sensitive to sex differences in associations of nature exposure with brain and cognitive health, perhaps simply because individuals may be more likely to interact with or be impacted by nature spaces closer to their residence. Indeed, 300 m is commonly used in health research to reflect the immediate residential area and in policy-making as the maximum recommended distance to public open spaces[69,70]. Despite this, previous studies have found significant associations of nature exposure with brain and cognitive health across buffer sizes varying between 100 and 2000 m, and further work is required to clarify if any buffer sizes may be most relevant to brain or cognitive outcomes[3,44].

There are many biological and behavioural mechanisms by which sex could interact with nature–brain health pathways to elicit sex differences in brain or cognitive outcomes. Examples include sex differences in stress responses[31,32,35,37], mental health[23], air pollution effects[29,30], and physical activity responses[71–73], as well as differences in the neuroprotective effects[74] and aging-related trajectories[25,75] of sex hormones. In particular, we note that the age of the UK Biobank sample overlaps with the typical age range of perimenopause and early menopause in females. The negative neuroendocrine changes during the menopausal transition[25,75] may attenuate the possible beneficial effects of nature exposure in females, relative to males. Other notable factors exhibiting sex differences in the current study include depression, which was more prevalent in females, and cardiovascular problems (predominantly high blood pressure), which were more prevalent in males. The sex differences in these dementia risk factors[58] could modulate either sex's sensitivity to the potential benefits of nature exposure. These potential pathways for sex differences are an important avenue for future investigation.

This study had several limitations. First, the cross-sectional nature of the analysis precludes causal inference and allows for reverse-causality (e.g., healthier people choosing to live closer to nature)[22,76]. Second, the binary, land cover-based residential nature exposure measure does not account for the quality of the natural environment, the participant's actual engagement with the nature, and the participant's nature exposure during daily activities beyond the residence[22]. Third, the study sample was cognitively healthy, 98% White/Caucasian, and based in Great Britain, so results may not be generalizable to those with cognitive or neurological impairment (e.g., dementia or stroke), or to other ethnicities or other geographical regions. Fourth, these analyses were exploratory, with no adjustment for type I error inflation from multiple comparions[77]. Finally, although the APOE ε4 genotype is a critical dementia risk factor[58], we were unable to include it as a covariate in our models because we did not have access to the necessary genotyping data.

Future longitudinal analyses examining associations between nature exposure and changes in brain health will help verify our findings and provide stronger evidence for a potentially causal relationship[76]. Alongside observational research, experimental studies, such as randomized controlled trials, are required to provide causal evidence and to investigate the effects of various types and doses of nature exposure on brain and cognitive health[55,78]. Future research should also test other potential moderators (e.g., age, menopausal status, ethnicity), as well as potential mediators (e.g., physical activity, stress, mental health, air pollution) of nature exposure on brain health[3,22].

Among the strengths of our study, this was one of very few studies assessing—and one of even fewer studies statistically testing for—sex differences in the association of nature exposure with brain and cognitive health. Also, to our knowledge, we used by far the largest sample size among the few studies assessing the association of nature exposure with brain structure[3,6,41].

Conclusions

Greater residential nature exposure may be associated with better brain and cognitive health in adults, and males may benefit more than females. These cross-sectional results add to the emerging observational literature on the potential utility of nature spaces as a community-level dementia prevention strategy. Future longitudinal studies are needed to verify and extend these findings.

Data availability

We analyzed data extracted from the UK Biobank on October 7, 2021 as part of UK Biobank Application 69022. Access to UK Biobank data is not publicly available but can be accessed by applying to the Research Analysis Platform described at the following link: https://dnanexus.gitbook.io/uk-biobank-rap. Code for all analyses can be viewed at: github.com/noseDr/UK-Biobank-Nature-Exposure-and-Brain-Health/tree/main.

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Author contributions
Conception: M.N., R.S.F., T.L.A. Data acquisition: L.A.G. Data analysis: R.S.F., M.N. Interpretation of results: M.N., R.S.F., T.L.A, L.A.G., T.C.H. Manuscript writing: M.N., R.S.F., T.L.A. Manuscript revisions: M.N., R.S.F., T.L.A, L.A.G., T.C.H.

Declarations

Competing interests
The authors declare no competing interests.

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